

ANDRE: AN ATTENTION-BASED NEURO-SYMBOLIC DIFFERENTIABLE RULE EXTRACTOR FOR INDUCTIVE LOGIC PROGRAMMING

Iman Sharifi¹, Peng Wei¹, and Saber Fallah²

¹Dept. of Mechanical and Aerospace Engineering, George Washington University, USA

²Dept. of Mechanical Engineering Sciences, University of Surrey, UK
 {i.sharifi,pwei}@gwu.edu, s.fallah@surrey.ac.uk

ABSTRACT

Inductive Logic Programming (ILP) aims to learn interpretable first-order rules from data, but existing symbolic and neuro-symbolic approaches struggle to scale to noisy and probabilistic settings. Classical ILP relies on discrete combinatorial rule search and is brittle under uncertainty, while differentiable ILP methods typically depend on predefined rule templates or inaccurate fuzzy operators that suffer from vanishing gradients or poor approximation of logical structure when reasoning over probabilistic predicate valuations. This paper proposes an *Attention-based Neuro-symbolic Differentiable Rule Extractor (ANDRE)*, a novel ILP framework that learns first-order logic programs by optimizing over a continuous rule space with attention-based logical operators. ANDRE replaces both rule templates and logical operators with fully differentiable, attention-driven conjunction and disjunction operators that approximate logical min–max semantics, enabling accurate, stable, and interpretable reasoning over probabilistic data. By softly selecting, negating, or excluding predicates within each rule, ANDRE supports flexible rule induction while preserving symbolic structure. Extensive experiments on classical ILP benchmarks, large-scale knowledge bases, and synthetic datasets with probabilistic predicates and noisy supervision demonstrate that ANDRE achieves competitive or superior predictive performance while reliably recovering correct symbolic rules under uncertainty. In particular, ANDRE remains robust to moderate label noise, substantially outperforming existing differentiable ILP methods in both rule extraction quality and stability.

1 INTRODUCTION

Symbolic artificial intelligence (AI) has long provided a foundation for interpretable reasoning, enabling systems to derive logical conclusions from structured knowledge (Colelough & Regli, 2025; Garcez & Lamb, 2023). A prominent approach within this paradigm is Inductive Logic Programming (ILP), which combines machine learning with first-order logic to infer human-readable rules from examples (Muggleton et al., 2012). Given a set of examples and background knowledge, ILP aims to discover logical clauses that explain observed facts, making it an essential tool in knowledge-base systems, explainable AI (Socher et al., 2013; Yang et al., 2017), and even safety-critical domains such as autonomous driving (Sharifi et al., 2025) and aviation (Acharya et al., 2025).

Classical ILP methods—such as Metagol (Cropper & Muggleton, 2015; Muggleton et al., 2015; Inoue et al., 2009) and Popper (Cropper & Morel, 2021)—tackle the ILP problem using a variety of heuristic, search-based strategies. While these approaches produce fully interpretable results, a major limitation lies in their sensitivity to noise in the data labels (Evans & Grefenstette, 2018). In addition, the rule discovery process typically requires considerable user intervention to define appropriate language bias—e.g., mode declarations and constraints—making the overall pipeline cumbersome and labor-intensive (Cropper & Dumančić, 2022). Furthermore, the lack of seamless

integration with neural networks restricts the scalability and adaptability of classical ILP methods in modern, large-scale machine learning applications.

To address these challenges, neuro-symbolic ILP methods have emerged, aiming to integrate ILP with neural networks to enhance scalability and robustness (Cropper & Dumančić, 2022). By leveraging differentiable optimization, these approaches formulate rule induction as a gradient-based learning problem, which facilitates more effective handling of noisy data and redundant predicates. Broadly, these methods fall into two categories: those that assume prior knowledge of the rules, and those that operate entirely without it. In the first category, rule induction is guided by candidate rules typically generated through general rule templates (Evans & Grefenstette, 2018) or beam search techniques (Shindo et al., 2021; 2023). Although these approaches can efficiently identify plausible rules, they often depend on domain-specific knowledge or involve ad hoc search procedures. Furthermore, due to their reliance on prior rule generation using templates and the manual conversion of background knowledge into readable valuations, these methods are not fully differentiable, limiting their compatibility with end-to-end neural network architectures (Evans & Grefenstette, 2018).

In contrast, the second category assumes no prior knowledge of candidate rules (Yang et al., 2017; Gao et al., 2022; 2024). Instead, these methods explore rules by assigning trainable weights to vector or matrix embeddings that represent potential predicate valuations. While these approaches are differentiable, partially robust to noise, and free from reliance on predefined rule templates, they mostly depend on soft, inaccurate approximations of logical conjunctions and disjunctions, using product-based t -norms and s -norms (Yang et al., 2017; De Raedt & Kersting, 2008) and regularized geometric mean (Wang & Pan, 2022), or trainable, less-interpretable parameterized function approximators (Badreddine et al., 2020; Sen et al., 2022). Although product-based operators are accurate for Boolean inputs (Sen et al., 2022; Payani & Fekri, 2019), their performance deteriorates with probabilistic data, where the product tends to rapidly decay toward zero (Hájek, 2001), leading to vanishing gradient issues (Shakarian et al., 2023). Parameterized approximators, on the other hand, often sacrifice interpretability by representing logical operations as complex nonlinear functions (Richardson & Domingos, 2006; Dong et al., 2019). To date, attention-based neuro-symbolic ILP approaches, such as NeuralLP (Yang et al., 2017) and DILR (Wang & Pan, 2022), have emerged to overcome the limitations of the existing approaches, integrating attention mechanisms (Vaswani et al., 2017) with logical operators. However, despite their effectiveness in general reasoning and question answering, their internal architectures still rely on classical logical operators, thereby inheriting the associated limitations. Given the limitations of existing neuro-symbolic ILP methods, there is a clear need for an alternative approach—one that extracts rules without predefined templates, avoids inefficient logical operators, and maintains both interpretability and structural flexibility.

This paper proposes a novel neuro-symbolic ILP method, called ***Attention-based Neuro-symbolic Differentiable Rule Extractor (ANDRE)***, which learns first-order logic programs from positive and negative examples using a fully differentiable and interpretable architecture. Operating without any assumptions about candidate rules or search-based heuristics, ANDRE reframes rule induction as learning over a continuous rule space in which each body predicate is softly selected, negated, or excluded using a novel fuzzy activation function. Furthermore, ANDRE employs softmin/softmax attention-based operators instead of classical logical operators, to effectively represent conjunction/disjunction operators. To the best of our knowledge, this is the first work to explicitly use softmin and softmax attention as a differentiable approximation of conjunction and disjunction in ILP. ANDRE’s novel attention-based operators are both accurate and fully differentiable. These modules are well-suited not only for Boolean data but also for continuous probabilistic data and mitigate vanishing gradient issues that typically hinder differentiable rule learners. Moreover, ANDRE is both scalable to large knowledge bases and strongly robust to noise in the data labels. The main contributions of this paper are:

- First, we introduce a new continuous rule space formulation for ILP, in which the discrete inclusion, negation, and exclusion of predicates are relaxed into differentiable probability distributions, enabling gradient-based learning of symbolic programs without rule templates.
- Second, we propose attention-based logical operators that provide a differentiable approximation to symbolic min–max conjunction and disjunction, allowing logical reasoning to be performed reliably over probabilistic predicate valuations rather than crisp Boolean facts.

- Third, we derive a set of syntactic regularization losses, with a loss hyperparameter scheduling technique, that enforce variable inclusion, connectivity, and digitization, ensuring that the learned programs correspond to valid and extractable first-order logic rules.
- Finally, through extensive experiments on classical ILP benchmarks, large-scale knowledge bases, and synthetic datasets with noisy supervision, we demonstrate that ANDRE achieves competitive or superior predictive performance while significantly improving rule extraction quality and robustness under uncertainty compared to existing differentiable ILP methods.

The remainder of the paper is organized as follows. Section 2 reviews the foundational concepts and necessary background. Section 3 introduces ANDRE’s methodology in detail. Section 4 presents experimental evaluations. Finally, Section 5 draws conclusions.

2 PRELIMINARIES

Inductive Logic Programming (ILP). As a subfield of machine learning, ILP focuses on learning logical rules from structured data. Given a background knowledge base $\mathbb{B}$, a set of positive examples E^+ , and a set of negative examples E^- , the objective of ILP is to learn a hypothesis H such that:

$$\forall e \in E^+ : \mathbb{B} \cup H \models e, \quad \forall e \in E^- : \mathbb{B} \cup H \not\models e, \quad (1)$$

where $\models$ denotes logical entailment. The hypothesis H comprises a set of first-order logic rules that best explain the observed examples while remaining consistent with the background knowledge. A typical ILP rule is a same-head disjunction composed of several subrules, defined as:

$$h(X_1^h, X_2^h, \dots, X_{n_h}^h) \leftarrow \bigvee_{i=1}^n h_i(X_1^h, X_2^h, \dots, X_{n_h}^h), \quad (2)$$

where h is the target head predicate, and the arguments $X_k^h \in \mathbb{R}$ are the head variables shared across all subrules. Inheriting all head variables from h , each subrule h_i is defined as a conjunction of specific body predicates:

$$h_i(X_1^h, X_2^h, \dots, X_{n_h}^h) \leftarrow \bigwedge_{j=1}^m B_{ij}(X_1^{b_j}, X_2^{b_j}, \dots, X_{n_{b_j}}^{b_j}), \quad (3)$$

where B_{ij} represents a potential body predicate in h_i , and $X_k^{b_j} \in \mathbb{R}$ are the variables associated with the j -th predicate. The arity n_{b_j} can vary depending on the predicate. The ILP objective is to identify the optimal assignment of B_{ij} in every subrule h_i , ultimately constructing the complete rule structure in Eq. 2. Due to the combinatorial nature of subrules, which involve various configurations of body predicates, optimizing them using neural networks—typically designed for fixed-size inputs—poses significant challenges.

Neuro-symbolic Inductive Logic Programming. Neuro-symbolic ILP aims to combine the symbolic reasoning capabilities of traditional ILP systems with the robustness and adaptability of neural networks. These methods reformulate the ILP task as a continuous, differentiable process, casting rule induction as an optimization problem in which logical rules are learned by minimizing a loss function over a dataset of N labeled examples $E^{N \times (m+1)}$. Each example $e \in [0, 1]^{(m+1)}$ includes the valuations of m potential body predicates, along with a corresponding binary output $h \in \{0, 1\}$. The goal is to map the inputs to outputs while discovering explicit logical connections between the body predicates and the target head predicate. This formulation enables training via gradient descent, allowing the model to generalize from ambiguous patterns and tolerate noise or imperfections in the data.

From a technical perspective, neuro-symbolic ILP typically maps positive and negative examples to binary labels, transforming the problem into a binary classification task that can be addressed using standard loss functions. The total loss function commonly consists of two main components: (1) *semantic loss* and (2) *syntactic loss*. Semantic loss—typically formulated as binary cross-entropy—guides the model in identifying high-level relationships between body predicates and the head predicate. It evaluates which subpredicates contribute most to reducing prediction error. Syntactic loss, in contrast, enforces structural correctness by ensuring that the learned rules conform to valid logical formats and general ILP constraints.